

To be completed by the professor			
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	Course: Computer Programming I		
	Program Year: 2026	Group: 1	Campus: Madrid
	Type of exam: ☒ Quiz ☐ Midterm ☐ Final ☐ Other: _____		
	Professor: Daniel Precioso		Date: 23/09/2026
	Total time allowed: 20 minutes		Num. Pages: 8

Student Name:	Student ID:
To be completed by the student	

DO NOT TURN THIS PAGE UNTIL THE PROFESSOR INSTRUCTS YOU TO DO SO.

EXAM RULES AND GUIDELINES

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Your ID, pens or pencils, erasers, and calculator (*only if approved by your professor*).

- All **electronic devices** (phones, smartwatches, headphones, etc.) must be **switched off and stored away**.
- **Food and drinks are not permitted** in the examination room.
- All **unauthorised materials** (including bags, notes, books, electronic devices, etc.) must be placed at the **front of the room**.
- If you are **unsure** whether an item is permitted, check with your professor before the examination begins.

During the Exam

- **Communication of any kind with other students is prohibited** (including talking, making gestures, passing notes, etc.).
- **Do not access any unauthorised materials or devices.**
- If you have a **question**, **raise your hand** (please note that this may not be allowed in some exams.)
- **Students are not permitted to leave and re-enter the examination room under any circumstances.** If you leave the room, you will not be permitted to return and continue the examination.
- **All students must remain in the examination room for at least the first 30 minutes.** A quiet and respectful environment must be maintained at all times.

After the Exam

- Ensure that **your name** is clearly written on your examination paper before submitting it.
- **Do not speak until you have left the examination room.**
- Once you have submitted your examination paper, remain silent to avoid disturbing other students.
- **Leave the examination room quietly and in an orderly manner** when instructed to do so by the professor.
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Student's signature _____

All examination papers and any additional materials used during the examination
MUST BE SECURELY STAPLED together before submission.

Instructions

Circle the correct answer for each question, as each question has **only one** correct answer. Provide reasoning for your answer in the space next to each question. If you need more space, use the blank page at the end of the exam.

A correct answer with a good explanation awards a point.

An incorrect answer, faulty or no reasoning, or no answer gives zero points.

Assume each code cell is **run in isolation**, meaning no other code has been run before it.

Questions

1. Which of the following statements is **FALSE** about numeric variables?
 - A. Dividing two integers using the division operator (/) always results in a floating value.
 - B. Integer and floating values can be used together in arithmetic operations.
 - C. Integer and floating values can be compared using relational operators.
 - D. Integer values use comma to separate thousands. Floating values use a dot to separate the integer part from the decimal part.

2. Which of the following lines will cause an error when executed?

```
1 x = 6 # Line 1
2 7 = y # Line 2
3 z = str(x) + str(y) # Line 3
```

- A. Line 2.
- B. None. All lines will execute without error.
- C. Line 1.
- D. Line 3.

3. Which of the following lines will cause an error when executed?

```
1 x = 2 ** 0.5 # Line 1
2 y = 3 ^ 4 # Line 2
3 z = "Hello, World!" * 10 # Line 3
```

- A. Line 3.
- B. Line 1.
- C. Line 2.
- D. None. All lines will execute without error.

4. The user runs this code and, when prompted, enters `int(21)` in the input field. What will happen?

```
1 age = input("Enter your age: ") # int(21)
2
3 if age >= 18:
4     print("You are an adult.")
5 else:
6     print("You are a minor.")
```

- A. The program prints "You are an adult."
- B. The program prints "You are a minor."
- C. The program errors.
- D. The program runs without any output.

5. After running the following code, what will be the value of `number`?

```
1 number = 3
2 exponent = 2
3
4 for e in range(exponent):
5     number = number * number
```

- A. `number` will be 9 (3^2).
- B. `number` will be 27 (3^3).
- C. `number` will be 81 (3^4).
- D. `number` will be 3 (3^1).

6. Which names will Python print when we run this code?

```
1 ls_kpop = ["BTS", "Blackpink", "Huntrix"]
2
3 for band in ls_kpop:
4     print(band)
5     if band == "Blackpink":
6         break
```

- A. "BTS", "Blackpink" and "Huntrix".
- B. "BTS" and "Blackpink".
- C. "BTS".
- D. The code will raise an error.

7. The code below contains several questionable choices, but only one line creates a problem that may affect the program later. Which line is it?

```
1 input = input("What is your name? ") # Line 1
2 capitalized = input.capitalize() # Line 2
3 print(f"Hello, ", capitalized) # Line 3
4 print("""It is a pleasure to meet you!""") # Line 4
```

- A. Line 2.
- B. Line 4.
- C. Line 3.
- D. Line 1.

8. Read the code below. What variable name only exists within the scope of the function?

```
1 def calculate_area(length, width):
2     area = length * width
3     return area
4
5 length = 5
6 height = 10
7 area = calculate_area(length, height)
```

- A. height
- B. area
- C. length
- D. width

9. What will happen when you run the code below?

```
1 def sqrt(number):  
2     return number ** 1/2  
3     result = sqrt(16)  
4     print(result)
```

- A. The program will print 4.0.
- B. The program will run without any output.
- C. The program will raise an error.
- D. The program will print 8.0.

10. Which one of the following statements about functions is **FALSE**?

- A. There can be multiple return statements in a function.
- B. Type hints are not mandatory.
- C. Functions can be defined inside other functions.
- D. Functions must always contain a return statement.

Cheatsheet

Given a number named `x`:

- `x / y` divides `x` by `y`.
- `x ** y` raises `x` to the power `y`.
- `x ^ y` is the bitwise XOR operator.

Given a string named `s`:

- `str(x)` converts `x` into a string.
- `int(s)` returns an integer. Fails if `s` has invalid characters.
- `s * n` repeats the string `s`, `n` times.
- `s.capitalize()` returns a copy of `s` with the first character in uppercase and the rest in lowercase.

Some useful built-in functions:

- `range(start, stop)` generates numbers from `start` until `stop`, not including `stop`.
- `input(prompt)` returns a string entered by the user. The optional `prompt` is printed before reading the input.
- `print(a, b, ...)` prints multiple values separated by a single space.

Functions and control flow:

- `return` exits a function immediately and optionally returns a value.
- `break` exits the nearest enclosing loop immediately.

Solutions

Question 1: False statement: “Integer values use comma to separate thousands. Floating values use a dot to separate the integer part from the decimal part.”

Python does not use commas as thousands separators inside one integer literal. For readability, underscores may be used instead, as in `1_000`. A comma separates expressions and can create a tuple.

Question 2: Line 2.

The statement `7 = y` attempts to assign a value to the literal `7`. An assignment target must be a valid name or another assignable object. This is a syntax error, so the cell cannot be executed.

Question 3: None. All lines execute without error.

The expression `2 ** 0.5` computes a square root, `3 ^ 4` performs the bitwise XOR operation on two integers, and multiplying a string by `10` repeats it ten times.

Question 4: The program errors.

The `input()` function always returns a string. Typing `int(21)` in the input field produces the literal string `"int(21)"`; Python does not evaluate it as code. Comparing that string with the integer `18` raises a `TypeError`.

Question 5: number is 81 (3^4).

The loop runs twice because `range(2)` produces `0` and `1`. The first iteration changes `number` from `3` to `9`; the second changes it from `9` to `81`.

Question 6: “BTS” and “Blackpink”.

Each name is printed before the condition is checked. When `band` is `"Blackpink"`, `break` ends the loop, so `"Huntrix"` is never reached.

Question 7: Line 1.

The assignment stores the entered name in a variable called `input`, replacing access to Python’s built-in `input()` function in the current scope. Lines 2–4 are unusual but valid; the shadowed function name is the choice that can cause a later problem.

Question 8: width.

`width` is a function parameter and exists only in the function’s local scope. The name `length` also exists globally, `height` is global, and `area` is assigned both locally and globally.

Question 9: The program runs without any output.

Running the cell defines `sqrt` but never calls it outside the function. The lines after `return` are indented inside the function and are unreachable because `return` exits the function immediately.

Question 10: False statement: “Functions must always contain a return statement.”

A function without an explicit **return** statement is valid and returns **None**. The other statements are true.